

Report of Project 3: Premixed Laminar Flames

Russ, Rachel

*School of Aeronautics and Astronautics
Purdue University
February 26, 2026*

1 Hydrogen-Air Flame

1.1 Iteratively Convergent Results - Q1(a)

The results are not iteratively convergent. Fig. 1 shows that this is the case. This is because not all of the residuals are below $1e-6$ after all the iterations. They are also not leveled off after 1000 iterations. Both combined shows that the results are not iteratively convergent.

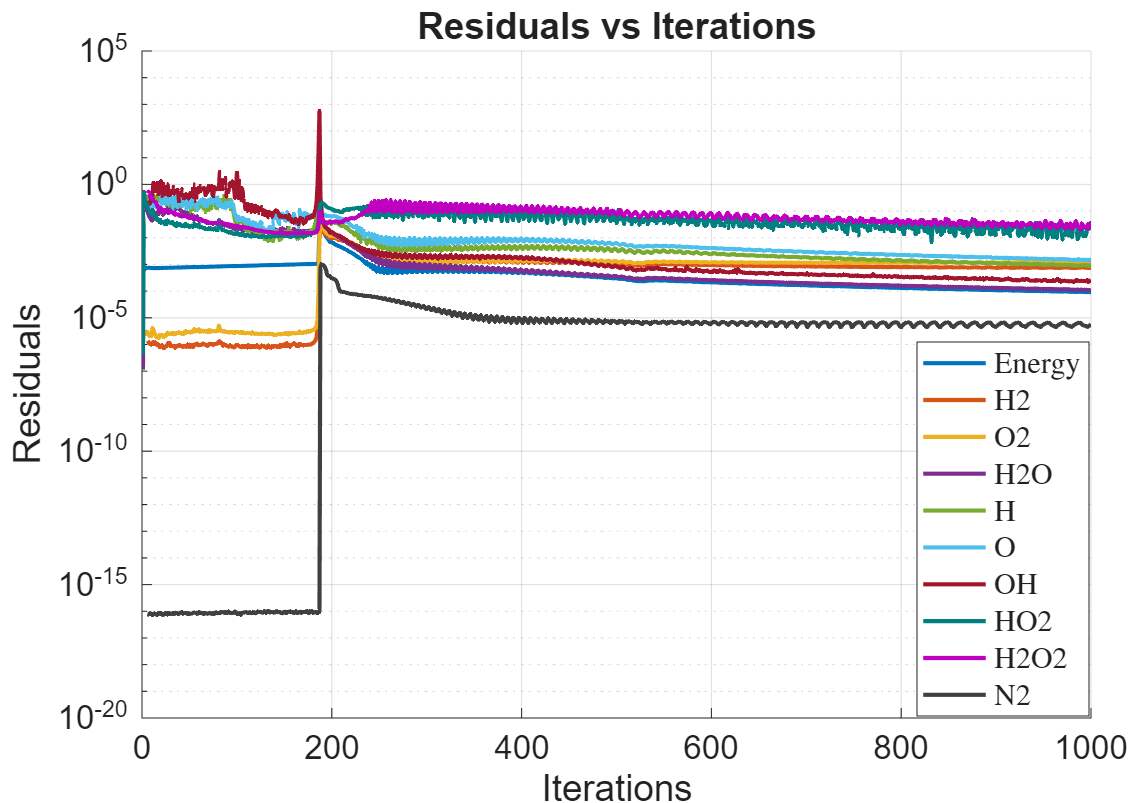


Figure 1 Residuals vs Iterations

However, it can be noted that the residuals have started to level off and become more linear, but after 1000 iterations this is not the case. In order to make the results iteratively convergent, there will have to be more iterations; however, that would cause the simulation to cost more and take more time.

1.2 Numerical Accuracy – Q1(b)

Some of the reasons that the solution is not numerically accurate is that the grid is not fine enough which would cause incorrect values. Especially near the outlet, the grid is very large. One reason for solving this problem is to make the grid finer. This would be more costly as it would require more solutions being solved. There is a tradeoff between the two.

Another reason that the solution may not be numerically accurate is that the convergence criteria is not small enough. If the solution does not converge, there is a good chance that the results are not accurate as they did not settle on a value.

One last reason could be because there are modeling oversimplifications. This could be that there is over assumption in the theory which is making the results incorrect. An example of this is that results are assuming 1-dimensional flow; however, the flame burner is not 1-dimensional. There are changes in the width of the burner which would cause the flow to be not 1-dimensional.

1.3 Grid Convergence – Q1(c)

The way that a grid convergence analysis is done in FLUENT is by refining the mesh. The results are run through a mesh of 3 different sized (small, medium, and course) to make the results independent of the mesh size. This way it can be seen how much the results vary due to the changes in mesh. The way to refine the mesh is done by the flow chart below in Eq.(1).

Adapt → Manual → Cell Registers → New → Create Region →
Choose Minimum and Maximum Values that Cover Whole Domain → Name it Entire Domain →
Display (1)

Eq.(2) shows how to create a domain so you can refine the mesh to make it finer.

Refinement Criterion → Cell Registers (Entire Domain) → Adapt (2)

Now all the cells are divided into four different cells to make a finer mesh. To evaluate at a courser grid, follow the flow below which is labeled as Eq.(3).

Coarsing Criterion → Cell Registers (Entire Domain) → Adapt (3)

This will make four cells one cell. Once the results are run, it is important to check how the solutions differ. If they are very similar, the results are grid independent.

1.4 Numerical Parameters that affect Numerical Accuracy – Q1(d)

Because subpart 1.3 makes the results grid-independent eliminating one source of error, it is important that the other sources of error are delt with. One way to do this is to look at the convergence criteria. The laminar flame speed (S_L) is calculated by Eq.4 below.

$$S_L = \frac{u_{burnt} \rho_{burnt}}{\rho_{unburnt}} \quad (4)$$

In Eq. 4, ρ_{burnt} is the density of the burnt part of the flow, in this case at a x value of 0, and $\rho_{unburnt}$ is the unburnt part of the flow, in this case, at a x value of 2 cm. u_{burnt} is the unburnt velocity of the flow, in this case, at a value of x at 2 cm. Fig. 2 shows the mass fraction of OH, maximum temperature, x-velocity, and laminar flame speed as functions of iterations. The results from two differing convergence criteria are plotted together.

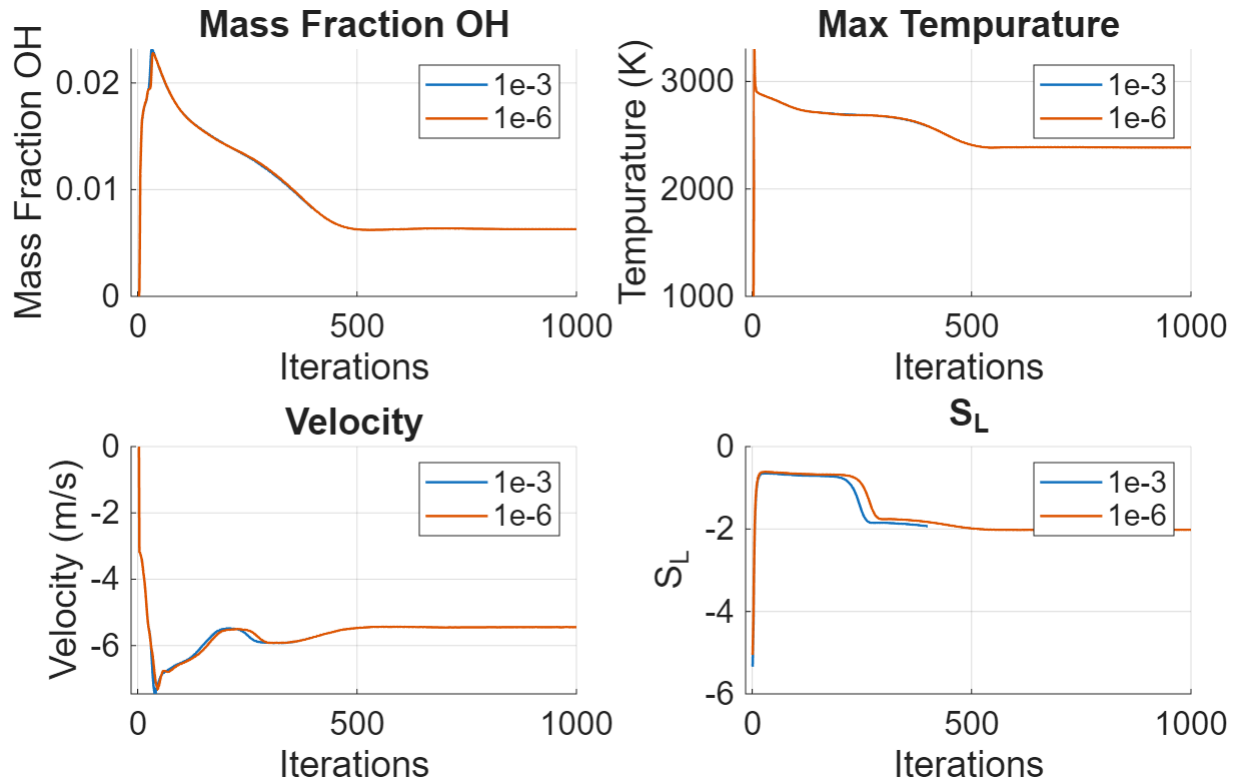


Figure 2 Numerical Parameters and Differing Convergence Criteria

All of these values are affected heavily by the convergence criteria. It can be shown that the difference between the accuracy of convergence criteria of 1e-3 and 1e-6 are not that different. They both overly each other for most of the graph. The only place that these two criteria differ is where the flame front is which make sense as this is where flame changes the most especially for velocity and laminar flame speed. However, even these differences are very small as they are almost unnoticeable on the graphs. Because a higher convergence criteria cost more in terms of iterations and time, from this point forward the rest of the data is conducted with a convergence criteria of 1e-3.

1.5 Affect of Thermal Diffusion on Transport Models – Q1(e)

Thermal diffusion is a very important parameter. This is because molecular diffusion is very important in premixed flame. Fig. 3 shows the mass fraction of OH, maximum temperature, velocity, and laminar flame speed. The laminar flame speed is calculated in the same way that it is calculated in section 1.4.

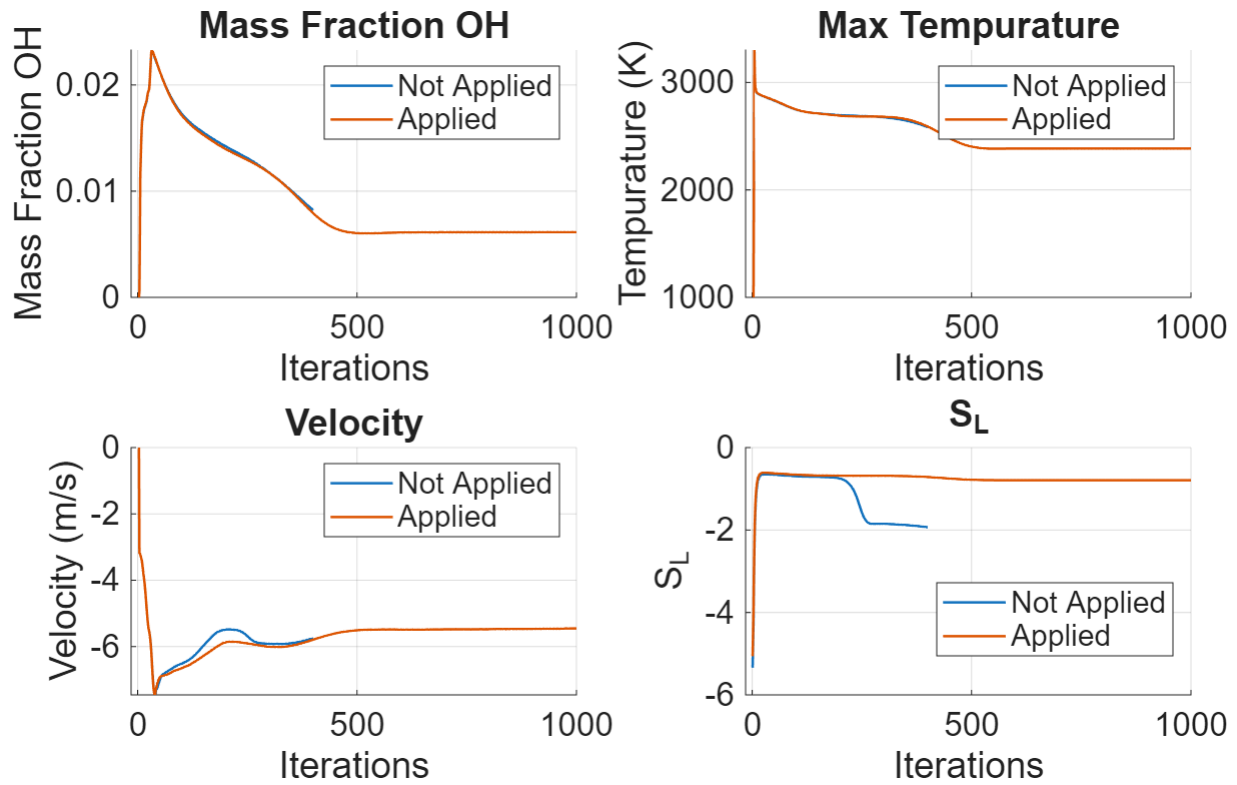


Figure 3 Thermal Diffusion Difference

It is evident from Fig. 3 above that the thermal diffusion model does not make much of a difference except for the laminar flame speed and velocity. For these, especially for laminar flame speed, it is evident that the affects of thermal diffusion make a difference.

We are using mixture average diffusion which is less accurate but more efficient. Full Multicomponent Diffusion is not used as it increases the cost and time it takes for the model to run.

1.6 Test Cases at Differing Flame Speeds – Q1(f)

The laminar flame speed changes based on how rich fuel or lean the flame is. The higher the equivalence ratio (ϕ), the more fuel rich the flame is. Due to the low temperature of 298 K, the flame is unburned. Table 1 shows the mass fractions used for each of the equivalence ratios.

Table 1 Mass Fractions of Reactants for Differing Equivalence Ratios

Equivalence Ratios	H ₂	O ₂	N ₂
$\phi = 0.8$	0.0228	0.2280	0.7492
$\phi = 1.0$	0.0285	0.2264	0.7451
$\phi = 1.2$	0.0337	0.2249	0.7414

For the stoichiometric case, the inlet velocity was set for -2.5 m/s. For fuel rich and fuel lean, the inlet velocity was set to -2 m/s and -1 m/s respectively. The negative value accounts for the direction of the flow. The laminar flame speed (S_L) can be calculated by Eq. (5) below.

$$S_L = \frac{\dot{m}}{A\rho} \quad (5)$$

In Eq. (5), where A is the cross-sectional area (depth is assumed to be 1 m). $\dot{m}$ is the mass flow rate, and ρ is the density of the gas. The cross sectional area is calculated based on the x-position and the equations that were originally given. This evaluation has the jump between equations at $x=0.5$ cm as this is where the two equations are equal. Eq.(6) is the equation for the strain rate (a).

$$a = \frac{2U}{L} \quad (6)$$

In Eq.(6), U is the x-velocity, and L is the length of the section of importance, which is 2 cm. The linear region of the laminar flame speed vs strain rate figures is then put as an equation of linear best fit using the MATLAB polyfit function. Fig. 4, 5, and 6 show this linear fit equation with the actual equation for each of the equivalence ratios.

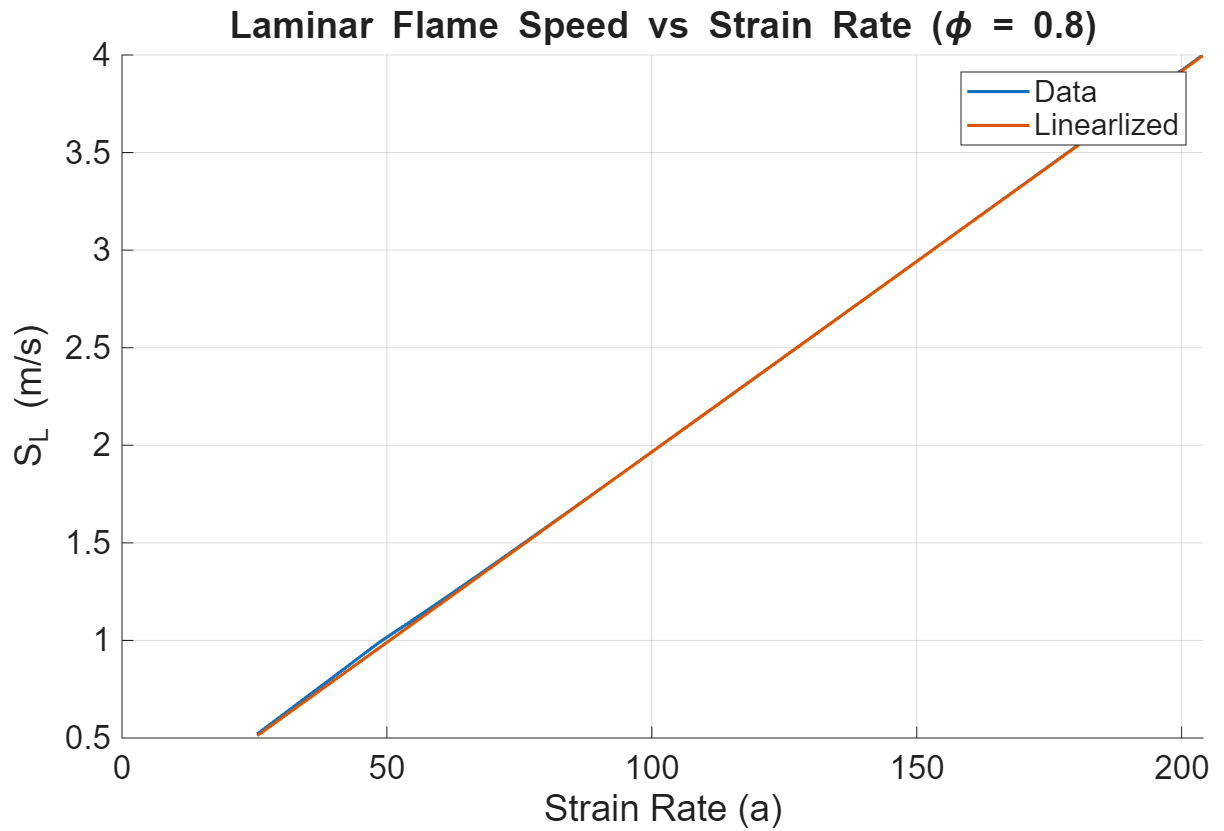


Figure 4 Laminar Flame Speed vs Strain Rate for Equivalence Ratio of 0.8

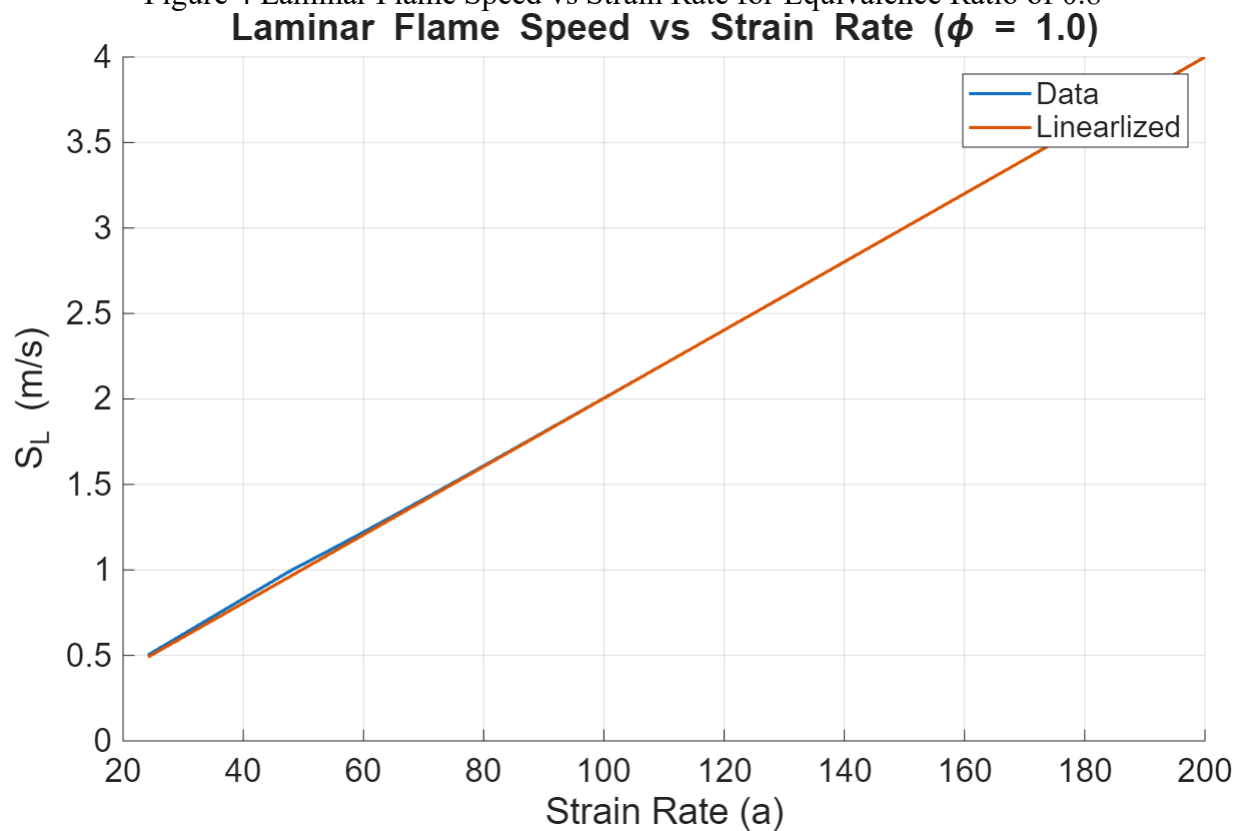


Figure 5 Laminar Flame Speed vs Strain Rate for Equivalence Ratio of 1.0

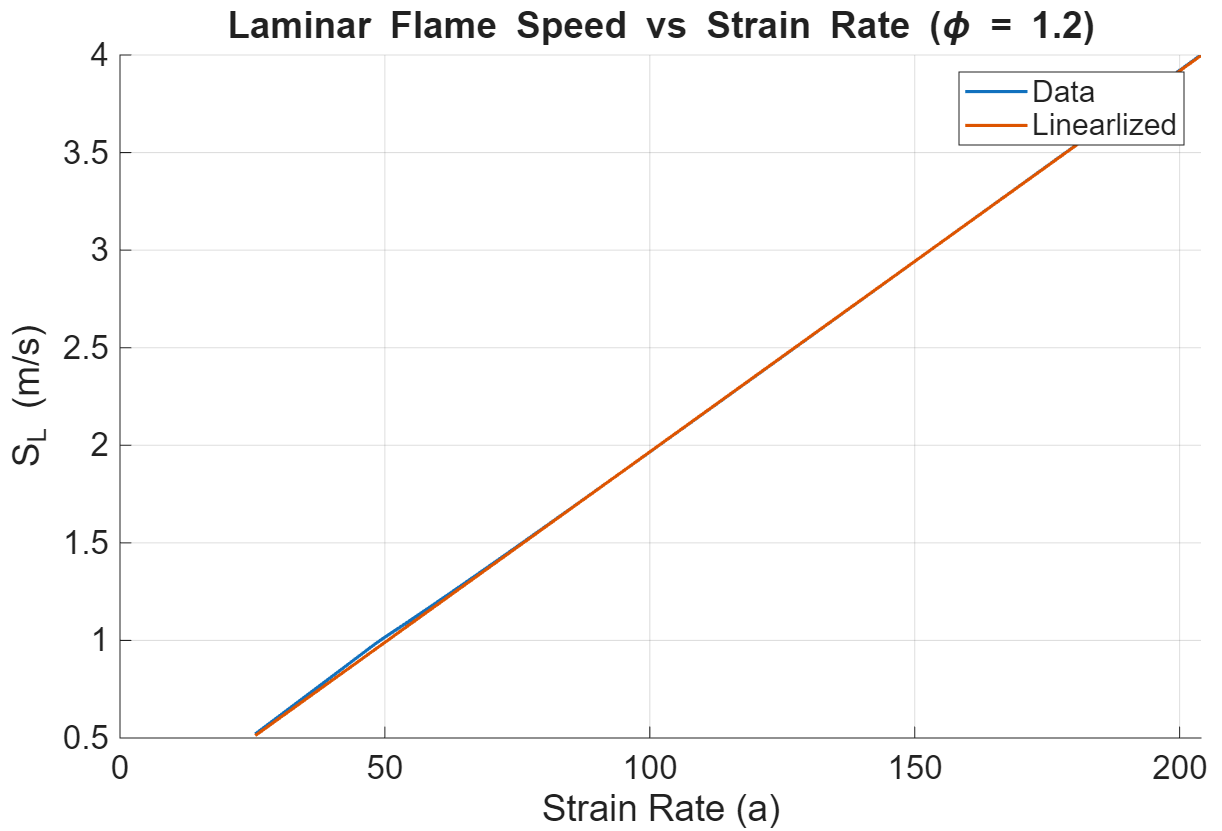


Figure 6 Laminar Flame Speed vs Strain Rate for Equivalence Ratio of 1.2

It can be shown by Fig. 4, 5, and 6 that the cases being unburned greatly simplifies the data and makes these results almost linear except for at low strain rates (low velocities). The y-intercept of this linear fit equation is the unstretched laminar flame speed. Fig. 7 below shows unstretched laminar flame speed as a function of equivalence ratio.

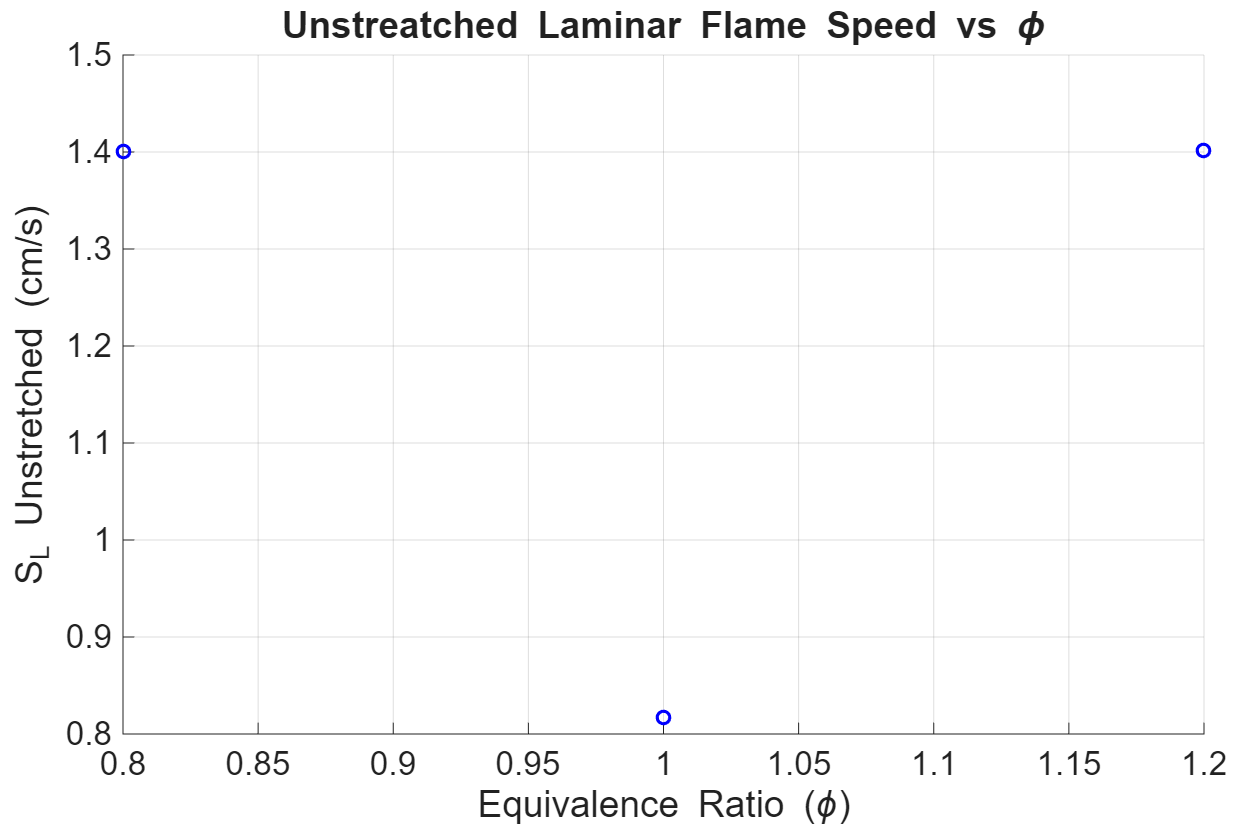


Figure 7 Unstretched Laminar Flame Speed as a Function of Equivalence Ratio
These values have a very interesting trend as the minimum is the stoichiometric condition and the greater values are for the fuel rich and fuel lean conditions.

1.7 Comparison with Experimental Results – Q1(g)

Fig. 8 below shows the experimental results of the laminar flame speed.

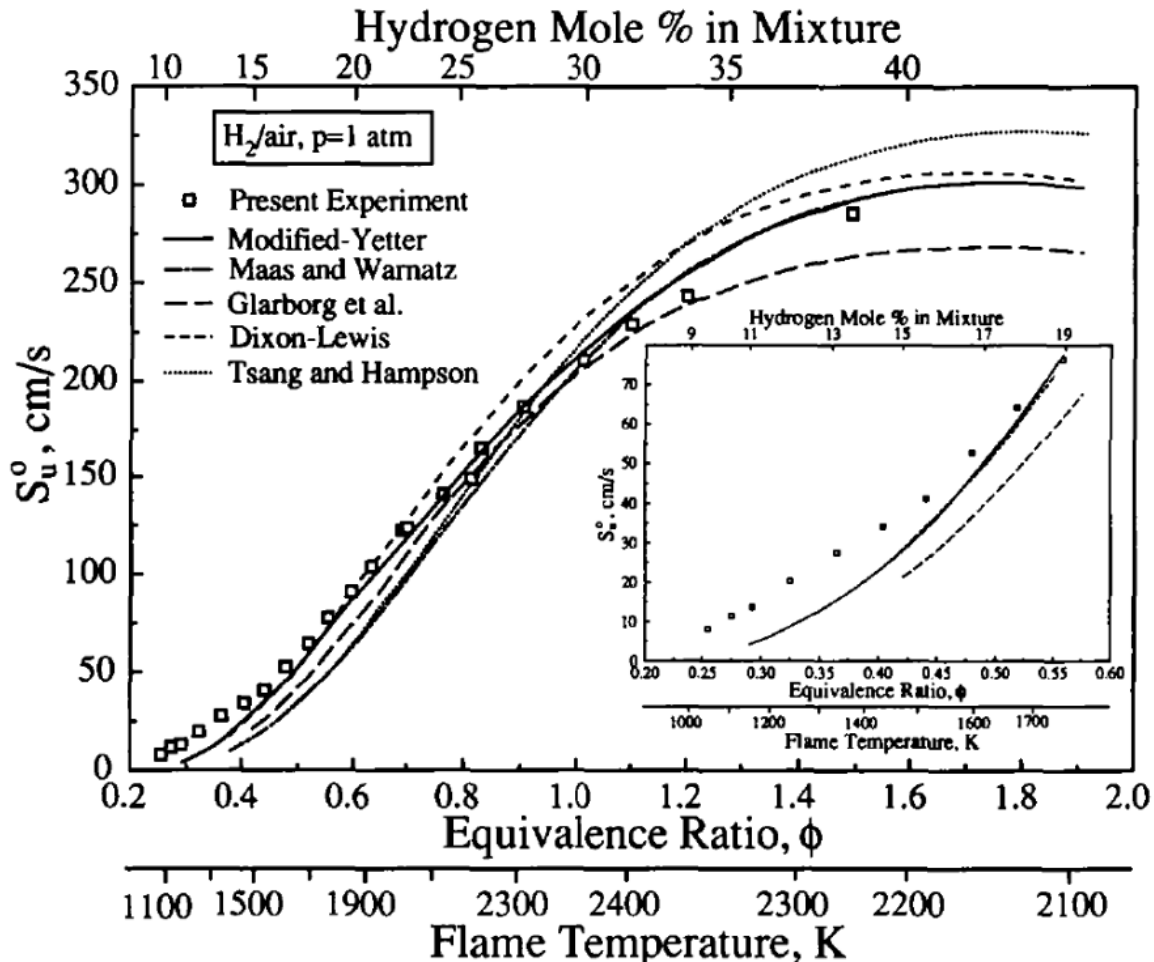


Figure 8 Experimental results of the laminar flame speed for hydrogen/air mixtures taken from Egolfopoulos & Law (Proc. Comb. Inst. 23, 333-340, 1990)

It is obvious that there is a directly proportional trend to the laminar flame speed and the equivalence ratio with the slopes of the lines greatest at an equivalence ratio of around 1.0. This coincides with the CFD data as the slopes of the laminar flame speed and the equivalence ratio is 1.95 cm/s for the fuel rich and fuel lean cases while it is 2.0 cm/s for the stoichiometric case. Roughly speaking, this matches up well with the experimental results. It is difficult to truly compare the results from FLUENT to the experimental results as there are not exact values given for each point.